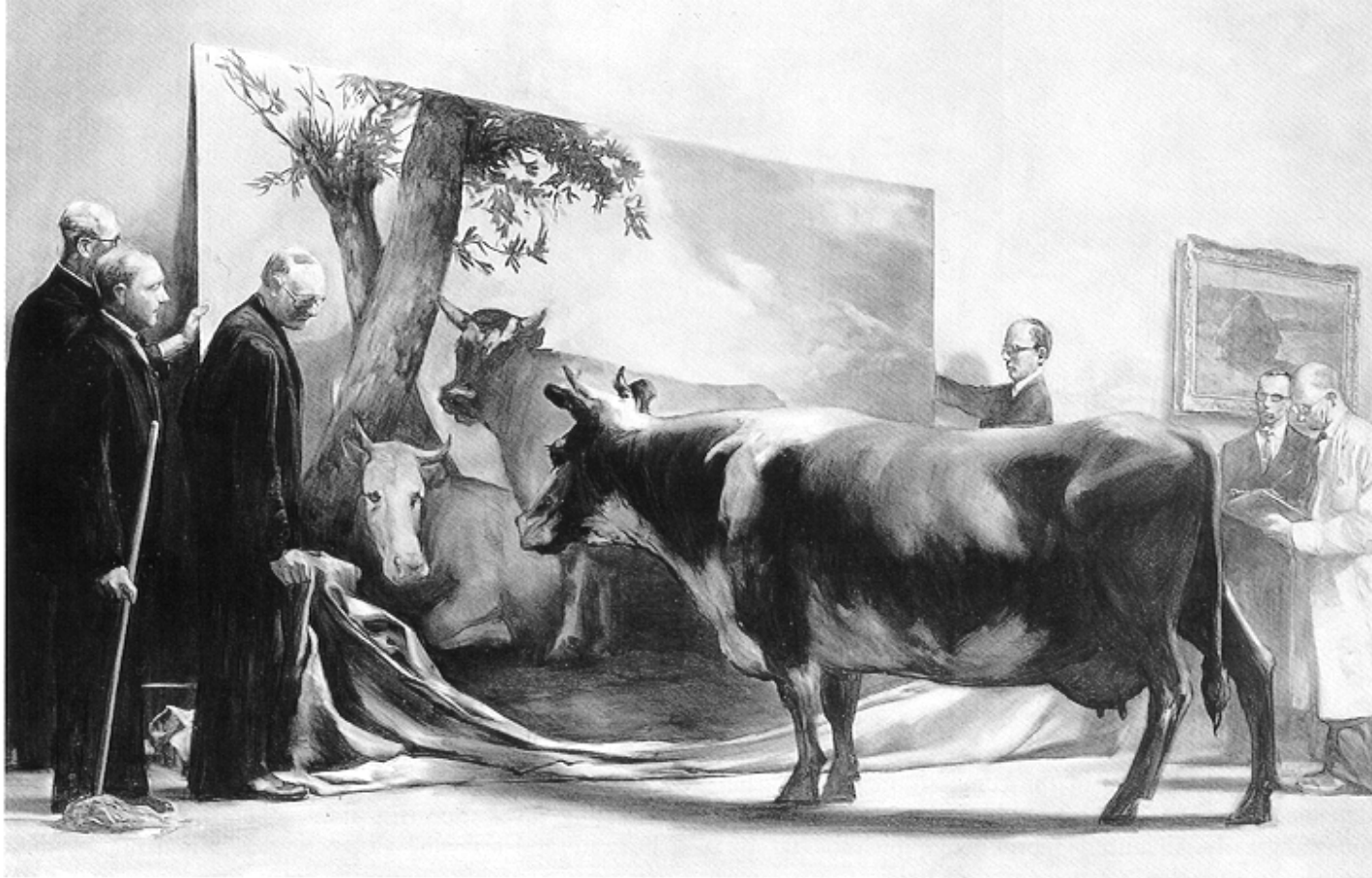


ENVX2001 – Applied Statistical Methods



Lecture 6 – MANOVA/PERMANOVA

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Outline

- Multivariate data and extending ecology
- Hypothesis testing and data exploration*
- Important techniques including
 - Cluster analysis*
 - nMDS
 - Monte Carlo approaches
 - PCA
 - MANOVA/PERMANOVA

This lecture

- MANOVA
 - Uses in ecology
 - Comparison to ANOVA
 - Process and theory underpinning it
 - Choosing a test statistic
 - Corrections for multiple comparisons?
- PERMANOVA

References

- Quinn and Keough (2002) – Chapter 16,18
- Quinn and Keough (2023) – Chapter 14,15,16
- Anderson, M. J. (2001). A new method for non-parametric multivariate analysis of variance. *Aust. Ecol.* **26**:32 – 46.
- Somerfield, P.J., et al. (2021). Analysis of similarities (ANOSIM) for 2-way layouts using a generalised ANOSIM statistic, with comparative notes on Permutational Multivariate Analysis of Variance (PERMANOVA). *Aust Ecol* **46**:911-26.

Take home messages

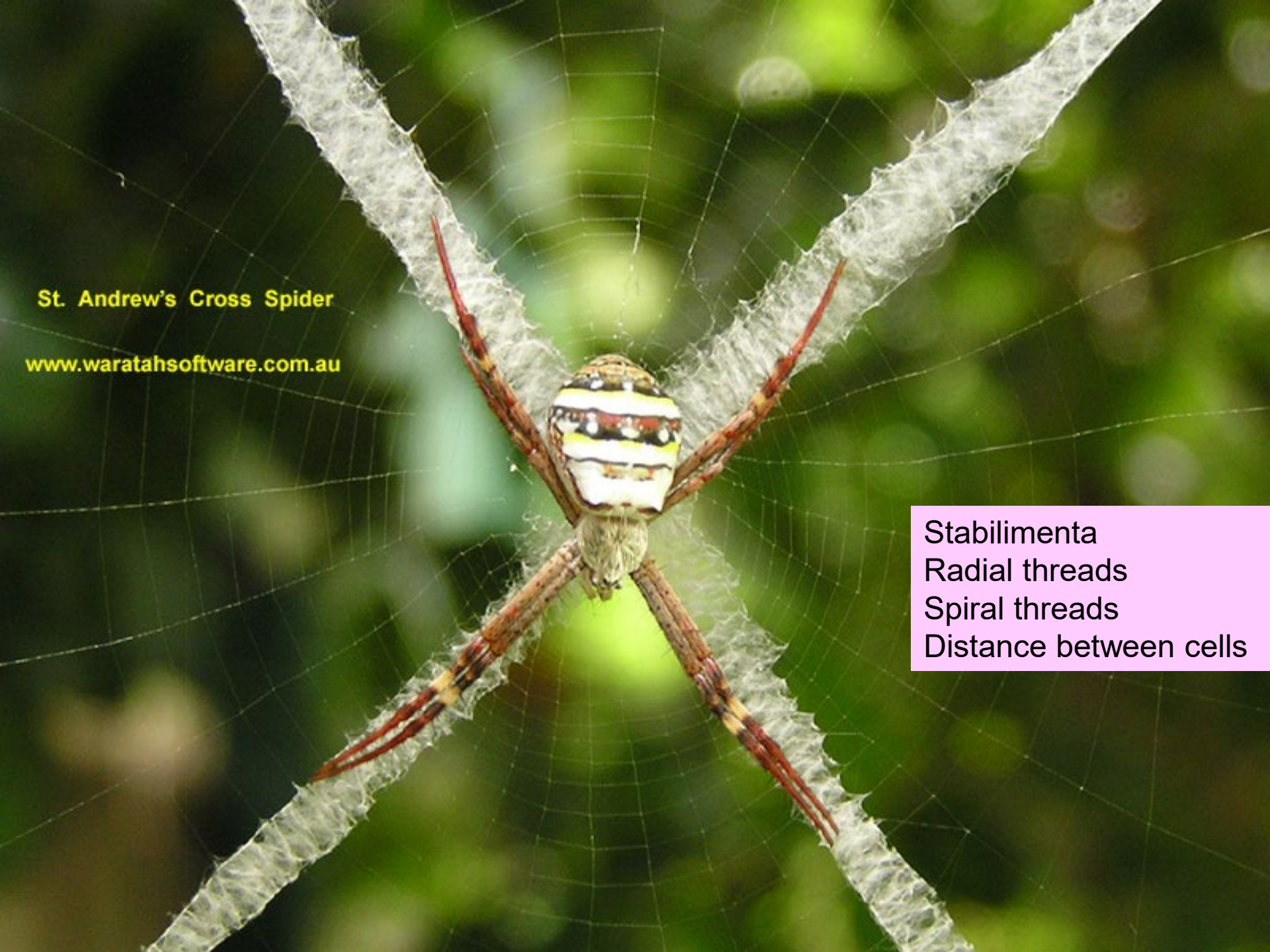
- Decision making and options
- Different approaches to MANOVA
 - Rationale
 - Pros and Cons
 - Interpretation of outputs
- Parallels to PERMANOVA



MANOVA - the question(s)



- Same as ANOVA - just on the linearly combined dependent variables instead just one dependent variable
- ***“Why the spider crossed the web”***
 - How St Andrew’s Cross spiders invest in their webs (relative to food, habitat etc.)
- ***“Plant fitness and restoration of degraded landscapes”***
 - Responses of target species to revegetation (with respect to how “good” they are after a certain period)
 - Use a phytometer to assess effects of landscape management

A close-up photograph of a St. Andrew's Cross Spider (Stilpnus argenteus) on its web. The spider has a yellow and black striped abdomen and long, thin, reddish-brown legs. It is positioned in the center of the web, which is a complex structure of spiral and radial threads. The background is a blurred green, suggesting foliage.

St. Andrew's Cross Spider

www.waratahsoftware.com.au

Stabilimenta
Radial threads
Spiral threads
Distance between cells



Stabilimenta
Radial threads
Spiral threads
Distance between cells

Prickly Parrot-pea *Dillwynia sieberi* (Fam. Fabaceae)

For a MANOVA?

Biomass

Mortality

Height

Flowers

Seed set







Prickly Parrot-pea *Dillwynia sieberi* (Fam. Fabaceae)



For a MANOVA?

- ↑ Biomass
- ↓ Mortality
- ↑ Height
- ↑ Flowers
- ↑ Seed set



(A) Benthic organisms in cores and food items in stomachs

	Point Judith	Ninigret	Green Hill
Benthic organisms in 5 cm core samples			
Polychaetes	81 (79)	99 (52)	61 (26)
Copepods	40 (38)	213 (160)	35 (22)
Crustaceans	1 (1)	1 (1)	6 (6)
Amphipods	1 (1)	6 (7)	10 (14)
Other	50 (38)	27 (13)	69 (47)
Total	174 (137)	346 (148)	180 (85)
No. of species	17 (7)	23 (3)	25 (4)
Food items in winter flounder stomachs			
Stomach fullness	63 (33)	80 (25)	58 (25)
Polychaetes	1.8 (1.5)	0.4 (0.7)	1.3 (1.5)
Copepods	1.8 (2.5)	0.1 (0.4)	0.7 (1.9)
Crustaceans	1.2 (1.7)	1.0 (2.2)	0.3 (0.9)
Amphipods	0.4 (0.7)	9.9 (9.5)	3.7 (4.2)
Other	1.3 (2.4)	0.2 (0.7)	0.6 (0.6)
No. of species	3.4 (2.2)	2.0 (1.0)	3.3 (1.6)

(B) MANOVA table for benthic organisms in cores

Source	df	Wilks' lambda	F	p
Vegetation	6	0.013	12.78	0.21
In/out	6	0.007	23.26	0.16
Vegetation*In/out	6	0.034	4.81	0.34

(C) MANOVA table for stomach contents

Source	df	Wilks' lambda	F	p
Vegetation	7	0.245	2.64	0.13
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From Menge et al. 2000
Winter flounder growth rates
and habitat quality
Mar Ecol Prog Ser
201: 287–299



Why MANOVA?

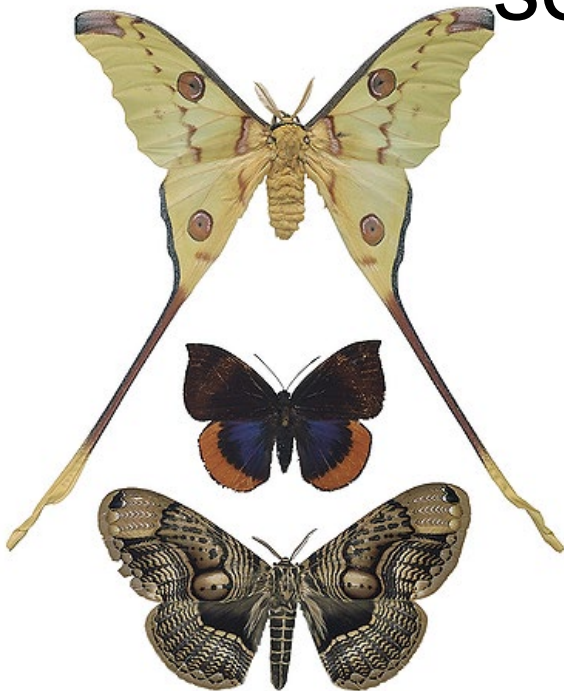
- Separate ANOVAS ignore relationships between dependent variables (lose information about correlations)
- Multiple univariate tests inflates overall type I error rate
- Effects along multiple c.f. single dimensions
- *Combination of DVs gives MANOVA greater power to detect differences*
- BUT... you need a good theoretical basis for combining variables.
- Don't include things just because you measured them

MANOVA - the process/meaning

- Multivariate extension of t-test and univariate ANOVA
- Extension of ANOVA -main effects and interactions are assessed on a combination of dependent variables
- MANOVA tests whether mean differences among groups on a combination of dependent variables is likely to occur by chance
- Similar to ANOVA in interpretation
 - Test statistics (note the choices here)
 - post-hoc tests
- Similar theory underpins it
- Shortcomings?



So what do we need to do
to critically evaluate
someone's ANOVA
output?



Basic requirements

- 2 or more dependent variables (continuous or ratio)
- 1 or more categorical independent variables (nominal or ordinal)

Kinds of research questions asked by MANOVA

- Same as ANOVA just on the linearly combined dependent variables instead just one dependent variable (maximizes the difference between groups)
- In factorial designs a different linear combination of the dependent variables is created for each main effect and interaction that *maximizes the group difference* separately



Are there any main effects?

Holding all other effects constant

Is a difference among groups greater than expected by chance?



Which things are most important?

- Which dependent variables are most important?
 - For significant main effects or interactions, on which individual dependent variable is there the most change (difference) “caused” by the levels of the independent variable?
Follow a significant MANOVA with individual ANOVAs in order to see extent of effects on dependent variables
- Which levels of the independent variable are significantly different?
 - If there are significant main effects on independent variables with more than two levels than you need to test which levels are different from each other
 - If there are interactions the interactions need to be taken apart so that the specific causes of the interaction can be uncovered

Assumptions of MANOVA

Assumptions - MANOVA

- **Independence** – observations should be statistically independent
- **Random sampling** – data should be randomly sampled from population of interest
- **Multivariate normality** – DVs collectively have multivariate normality within groups
- **Homogeneity of covariance matrices** – same as for ANOVA plus correlations between any DVs is the same in each group

Fundamental Considerations

- Interpretation of MANOVA results always in the context of the research design
- Smoke and mirrors - clever statistics and fancy graphics won't not make up for poor design



Missing data, unequal samples, number of subjects and power

- Missing data needs to be handled in the usual ways
- Unequal samples cause non-orthogonality - total sums of squares is less than all of the effects and error added up
- Power in MANOVA depends on the relationships among the dependent variables
 - highly correlated dependent variables weaken the power of the analysis
- PERMANOVA as a nonparametric option?



Multivariate normality

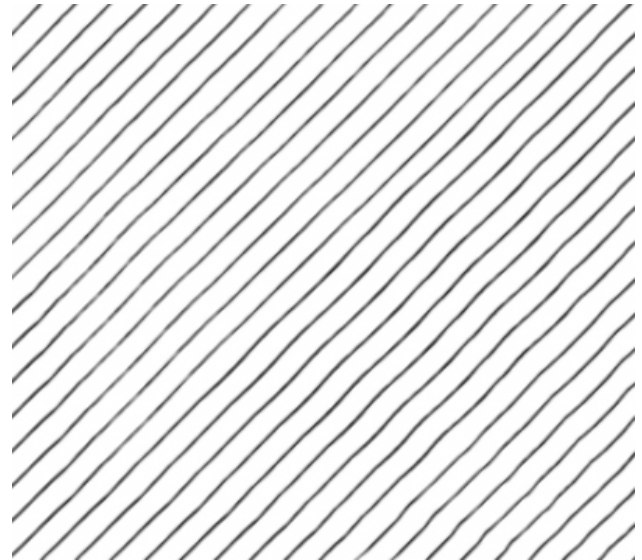
- assumes that the means of the various dependent variables in each cell and all linear combinations of them are normally distributed
- Difficult to show explicitly
- **In practice - if individual variables are normal, then they should have MVN**
- Key is to assess univariate normality
 - Plot your data – look for skewness, kurtosis, outliers, i.e., seek symmetry
 - Shapiro-Wilks test

Variance?

- **Homogeneity of covariance matrices**—same as for ANOVA plus correlations between any DVs is the same in each group
- Levene's test (for each DV)
- Box's test (to test the variance-covariance matrices)
 - Susceptible to non-normal data

Linearity

- MANOVA assumes linear relationships between all dependent variables
- Deviations from linearity reduce the power of the test because the linear combination of dependent variables does not maximize the difference between the groups



Different Multivariate test criteria

- Hotelling's Trace
- Wilk's Lambda
- Pillai's Trace
- Roy's Largest Root



Different Multivariate test criteria

- When there are only two levels for an effect all of the tests should be identical
- When there are more than two levels the tests should be nearly identical - not always the case
- Wilk's Lambda, Hotelling's Trace and Pillai's trace all pool the variance from all the dimensions to create the test statistic
 - E is error and H is hypothesized effect
- Roy's largest root only uses the variance from the dimension that separates the groups most (the first canonical root, the largest "root" or difference)

Different Multivariate test criteria

- Wilk's Lambda - $|E| / |H + E|$
- the ratio of error to effect plus error
 - A bit conservative - middle of the road power

<http://www.york.ac.uk/depts/maths/histstat/people/wilks.gif>



Different Multivariate test criteria

Hotelling's trace – $T (H/E)=C$

- Sum of the eigenvalues for each variate
- Look up C in a table to get the F value (analogous to an F-test)
- Very liberal test

Pillai's trace – $V = (H/(H + E))$

- Proportion of explained variation for each of the discriminant functions
- Very conservative

Roy's Maximum Root

- Linear combination of the observations that maximizes the F-ratio (between group variance while minimizing the within-group variance)
- Variable in terms of conservatism
- Susceptible to violations



<http://www.york.ac.uk/depts/maths/histstat/people/hotelling.gif>

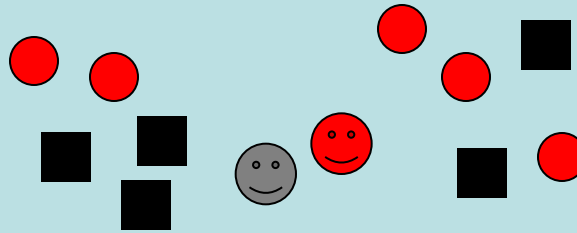
How to do a MANOVA in R?

```
data(iris)
# Need to bind the variables together
Y <- cbind(iris$Sepal.Length,
iris$Sepal.Width, iris$Petal.Length,
iris$Petal.Width)
model <- manova(Y ~ Species, data =
iris)
Summary(model)
```

PERMANOVA

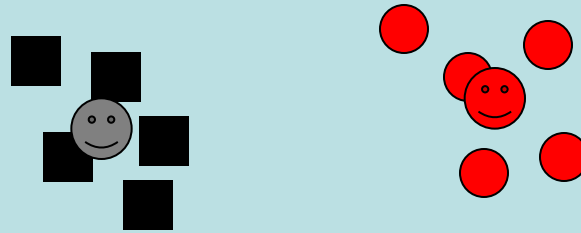
- works with any distance measure that is appropriate to the data, and uses permutations to make it distribution free
- Theoretically
 - Generate the distance matrix (usual choices)
 - Run the PERMANOVA using the same models as for MANOVA
- In R package *vegan* called *adonis*

Distances from centroids



Pseudo F based on
 $SS_{\text{between}} / SS_{\text{within}}$

Distances from centroids



Pseudo F based on

$$SS_{\text{between}} / SS_{\text{within}}$$

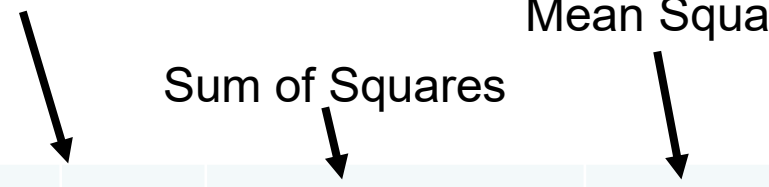
P values by permutations

PERMANOVA OUTPUT

Degrees of Freedom

Mean Square

Sum of Squares



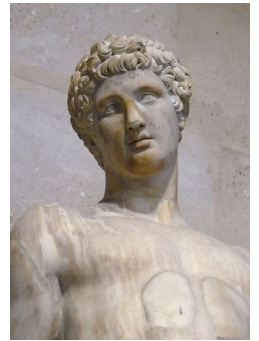
Source	df	SS	MS	Pseudo-F	P(perm)	perms
Co	4	28120	7030	2.6918	0.001	999
Sa	1	1919.4	1919.4	0.73495	0.74	999
CoxSa	4	6160.1	1540	0.58968	0.992	997
Res	66	1.72E+05	2611.6			
Total	75	2.09E+05				

How do you do a PERMANOVA in R?

#It is in the package vegan and
is called 'adonis'

```
adonis2(TransAntData  
~Community*Sample, data =  
AntFactors, permutations = 999,  
by = "terms")
```

#Code for pairwise tests given
in practical notes



Which do you choose?

- All fairly robust to violations of normality
- For the most part the most widely used is Wilk's lambda
- Can use Hotelling's Trace if
 - Manipulated (experimental) variables
 - Very clean design with no internal validity problems
- Pillai's trace is the most conservative so but if your design/data have problems (e.g. unbalanced, assumption violation, etc) Pillai's is supposed to be robust to these problems
- PERMANOVA – use it for your dissimilarities

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Follow-up ANOVA

- If MANOVA is significant, undertake separate ANOVA on each of the DVs
- Type 1 errors?
- Follow-ups “protected” by MANOVA (i.e. would produce a non-significant result)
- Still problematic – use corrections?



Assessing dependent variables

- The overall alpha level should be controlled for considering the multiple tests

$$\alpha_{overall} = 1 - (1 - \alpha_1)(1 - \alpha_2) \dots (1 - \alpha_p)$$

- The alpha levels can be divided equally or they can be set up to give more important tests a more liberal alpha level.

Bonferroni corrections and multiple comparisons?

- Experiment –wide error rates?
- Corrections involve dividing alpha by n (or sequentially reducing n)
- Using a p value other than 0.05 to claim that there is a difference at $\alpha = 0.05$
- E.g. for claim that there is a difference
 - Comparison 1 - $0.05/4=0.0125$
 - Comparison 2 - $0.05/3=0.017$
 - Comparison 3 - $0.05/2=0.025$
 - Comparison 4 - $0.05/1=0.05$



<http://www.york.ac.uk/depts/maths/histstat/people/bonferroni.gif>

MANOVA – a summary

- Make sure DVs are meaningful
- MANOVA
 - Assumptions
 - Relationships among DVs
 - Choice of test statistic
- Follow-up ANOVAs
- Consider PERMANOVA as an alternative
 - same interpretation with fewer assumptions